Discussion

Safety Implications

The ultimate motivation of our research is the theory that reverse engineering neural networks might help us be confident in their safety. Our work is only a very preliminary step towards that goal, but it it does begin to approach several safety-relevant issues:

Phase changes: If neural network behavior discontinuously changes from one scale to the next, this makes it more challenging for researchers and society to prepare for future problems.

In-Context Learning: In-context learning has been a topic of concerned speculation in the safety community. With less-capable neural networks, one might be tempted to treat their behavior as relatively fixed after training. (That said, demonstrations of adversarial reprogramming [21] shed some doubt on this assumption.) In-context learning highlights that model behavior can in some sense "change" during inference, without further training. Even if we think of in-context learning as "locating" an already-learned behavior [22], rather than learning something new, the behavior could be a surprising and unwanted off-distribution generalization.

Mesa-Optimization: There have been some concerns that the underlying mechanism of in-context learning might be mesa-optimization [14], a hypothesized situation where models develop an internal optimization algorithm. Our work suggests that the primary mechanism of in-context learning, at least in small models, is induction heads. We did not observe any evidence of mesa-optimizers.

Linking Learning Dynamics, Scaling Laws, and Mechanistic Interpretability

The in-context-learning phase change may be a useful "Rosetta stone" linking mechanistic interpretability, learning dynamics [23], and statistical physics-like empirical properties of neural networks (e.g. scaling laws or phase changes). If one wants to investigate the intersections of these lines of work, the phase change seems like an ideal starting point: a concrete example where these lines of inquiry are intertwined, which can be explored in small models, bounded in a small sliver of the training process, and is linked to a capability (in-context learning) the community is excited about.

Related Work

The general approach of this paper to reverse engineering transformers is based heavily on our previous paper, *A Mathematical Framework for Transformer Circuits*. There is much to be said about how that framework relates to other work in interpretability. Rather than repeating it, we refer readers to Related Work in our previous paper, especially discussion of the relationship to circuits [6], to analysis of attention heads (*e.g.* [24, 25, 26, 27, 28]), and to related mathematical analysis (*e.g.* [29]).

Building on that perspective, we here focus on how aspects of this paper raise new connections to the machine learning literature, separate from the connections raised simply by the underlying framework.

IN-CONTEXT LEARNING

Emergent in-context learning was compellingly demonstrated in GPT-3 [1]. A number of papers have studied how to effectively leverage in-context learning, especially with "prompt engineering" [11, 12]. But of particular importance to us, several papers have tried to study how and when in-context learning occurs (*e.g.* [13, 30, 31, 32]).

Some of the findings of these papers are consistent with the induction head hypothesis, or support our methodology:

- Kaplan *et al.* [13] is the origin of our approach for using loss at different token indices as a formalism for studying in-context learning.
- O'Connor & Andreas [30] find that preserving word order in contexts is important, as the induction head hypothesis would suggest.

However, there are also places where experiments in these papers seem in tension with the induction head hypothesis:

- O'Connor & Andreas [30] have some experiments suggesting that removing all words except nouns can improve loss. This seems inconsistent with the induction head hypothesis. However, they only find this for experiments where they retrain the model on modified data. This seems both less directly related to our work (because we aim to study models trained on natural data) and subtle to interpret (because retraining models introduces the possibility of run-to-run loss variation, and the measured loss differences are small). The experiments where they don't retrain models on modified data seem consistent with the induction head hypothesis.
- Xie *et al.* [31] finds that LSTMs outperform Transformers when fit to synthetic data generated by a Hidden Markov Model (HHM) designed to isolate a particular theoretical model of in-context learning. We generally expect Transformers to outperform LSTMs at in-context learning on natural text (as seen in Kaplan *et al.* [13]), with induction heads as a major explanation. But in the case of the Xie *et al*. experiments (which don't use natural text), we suspect that the structure of the synthetic data doesn't benefit from Transformers, and that LSTMs are perhaps better at simulating HMMs.

Note that we are using a broader conception of "in-context learning", rather than something as specific as "few-shot learning". This is in contrast with Brown *et al*. [1], which describes that a language model "develops a broad set of skills and pattern recognition abilities. It then uses these abilities at inference time to rapidly adapt to or recognize the desired task," with examples of tasks such as few-digit addition and typo correction. In our conception of "in-context learning", we refer to all the ways that a model rapidly adapts to or recognizes what is going on in the context, even if "what is going on in the context" isn't well-conceived-of as multiple "shots" of some other specific repeated task.

SCALING LAWS

Over the last few years, the observation that machine learning models change in smooth, predictable ways described by scaling laws [13] has emerged as a useful tool for modelling the properties of models before training them.

The relationship between scaling laws and mechanistic interpretability might be seen as analogous to the relationship between thermodynamics and the physics of individual particles. For both thermodynamics and scaling laws, even though the underlying system is very complicated, we're able to find simple relationships between the variables – for thermodynamics: entropy, temperature, volume and pressure; for neural networks: loss, compute, parameters, and data. In contrast, mechanistic interpretability studies the individual circuits underlying our models, vaguely analogous to how one might carefully study individual particles in physics. In physics, these two layers of abstraction were bridged by statistical physics.

Can we bridge these two levels of abstraction in machine learning? The induction head phase change is the first time we're aware of a bridge. They give us a phenomenon at the level of macroscopic properties at loss which can be explained at the level of circuits and mechanistic interpretability.

In fact, induction heads may be able to explain previous observed exceptions to scaling laws. In our work, 1-layer transformers seem very different from deeper transformers. This was previously observed by Kaplan *et al.* [13] who found that 1-layer transformers do not follow the same scaling laws as larger transformers. It seems quite plausible that the reason why the scaling laws are different for one-layer models is that they don't have induction heads.

PHASE CHANGES & DISCONTINUOUS MODEL BEHAVIOR

In the previous section, we discussed how scaling laws describe smooth predictable relationships between model loss and properties like scale. However, a more recent set of results have made the situation seem more subtle. While models' losses often scale in predictable ways, there are cases where behavior is more complex:

- Brown *et al.* [1] find that, while aggregate loss scales predictably with model size, models' ability to perform specific tasks like arithmetic can change abruptly.
- Power *et al.* [8] observe a phenomenon they call "grokking" where models discontinuously jump from random chance to perfect generalization as they train.
- Double Descent [33] is a phenomenon where model performance first gets worse due to overfitting as one makes a model larger (the "classical" regime), but then gets better again past a certain point (the "modern" regime). Generalizations of double descent can occur with respect to parameter size, dataset size, or amount of training [34]. These phenomena are not discontinuous in loss, but they are surprising trend reversals, and perhaps discontinuous in derivatives.

For more general discussion of these phase change phenomena, see a recent blog post by Steinhardt [35].

The discontinuous phase change behavior we observe with induction heads is most analogous to Power *et al.* [8]'s "grokking", in that it occurs over the course of training. We think our main contribution to this literature is linking the changes we observe to the formation of induction heads and a parameter-level understanding of the circuits involved. As far as we know, induction heads are the first case where a mechanistic account has been provided for a phase change in machine learning.